MA342 Project 3

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1 Introduction

Modeling of physical systems often requires understanding how energy propagates across complex geometries. Here we investigate two classes of partial differential equations (PDEs): the parabolic Heat Equation and the hyperbolic Wave Equation. These equations are fundamental in engineering and physics; for instance, the heat equation allows us to predict how electronic components reach thermal equilibrium to avoid failure, and the wave equation is essential in acoustics and structural analysis for understanding how vibrations propagate through materials.

Following the computational frameworks established in Chapter 11 of Holder and Eichholz [1], we can model thermal diffusion in various shapes and mechanical oscillations, both under various boundary conditions. The specific geometry of interest for the heat study introduces unique boundary challenges compared to standard rectangular grids. Furthermore, we extend the standard wave model to include a damping term to account for realistic energy loss in physical materials as they resist motion due to internal friction or air resistance.

2 Mathematical Background

Both models for this project were expanded from one-dimensional examples into 2D space. These models are then functions of the x, y coordinates in space along with the time variable t. The output of the thermal diffusion model is temperature in °C. The output of the mechanical oscillation model is the disc's displacement in meters.

2.1 Thermal Diffusion (The Heat Equation)

The propagation of temperature $u(x, y, t)$ is governed by the first order PDE:

$$\frac{\partial u}{\partial t} = \alpha \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) \tag{1}$$

where $\alpha = \frac{k}{c\rho}$ is the thermal diffusivity involving the materials thermal conductivity k, specific heat c, and density ρ. This equation describes a dissipative process where temperature gradients smooth out over time.

2.2 Mechanical Oscillation (The Wave Equation)

The formation of the "wave equation" that represents the mechanical oscillation of a geometric surface is represented by the partial differential equation:

$$\frac{\partial^2 u}{\partial t^2} = \alpha(\nabla \cdot \nabla u) \tag{2}$$

where ∇ represents the partial differentiation with respect to each of the spatial coordinates in the model, and α represents the ratio between the horizontal forces at any point inside the surface and its density.

3 Model Methodology

Both models were not solved using exact solutions or by using a PDE solver but rather through a numerical approach using time stepping.

3.1 Discretization

3.1.1 The Heat Equation

In order to appropriately step through a solution, the variables x, y, and t need to be broken down into appropriately small increments to achieved the desired model resolution while maintaining model accuracy within a suitable run time. To do this, x and y were broken down using Δx and Δy while t used Δt. The Δt variable will be further explored in the stability section but the Δx and Δy variables were set to 0.05 cm for both a decent resolution and a semi fast run time.

To calculate the state of the model at the next time step, we use the values from the previous step and the state of the surroundings. The equation for the the heat transfer model is shown below.

$$\begin{aligned} u_{x,y,t+1} = u_{x,y,t} & \\ & + \frac{k\Delta t}{c\rho\Delta x^2}\left(u_{x+1,y,t} - 2u_{x,y,t} + u_{x-1,y,t}\right) \\ & + \frac{k\Delta t}{c\rho\Delta y^2}\left(u_{x,y+1,t} - 2u_{x,y,t} + u_{x,y-1,t}\right). \end{aligned} \tag{3}$$

3.1.2 The Wave Equation

Since our model focuses only on the vertical displacement of the surface, the horizontal forces at all points are assumed to be equal given that the horizontal position does not change over time. Hence, α in (2) will be treated as a constant.

Utilizing Newton's Method, the sum of forces at any particular point, we find

$$\rho\Delta x\Delta y\frac{\partial^2 u}{\partial t^2}(x,t) = F_{net}. \tag{4}$$

With Δx and Δy representing the mesh face dimensions (position increment defining the complexity of the x and y axis on the grid mesh). The grid mesh contains a 1 m circle as the unit circle.

We now have a second-order relation of the displacement with respect to time, and now need a representation for the sum of forces at all points on the grid mesh. When looking at a infinitely small piece of the mesh parallel to the XY plane, we can assume that the horizontal forces in all directions at this piece are equivalent as we assume the horizontal position of any particle does not change with respect to time. Assuming the length of this infinitely small particle has an equal θ angle from its center, we can define an equivalent horizontal force acting on both the left and right sides of the particle as such:

$$cos(\theta(x - \frac{\Delta x}{2}))\ T\ (x - \frac{\Delta x}{2}) = cos(\theta(x + \frac{\Delta x}{2}))\ T\ (x + \frac{\Delta x}{2}). \tag{5}$$

With T representing the horizontal tension on both sides of the infinitely small particle and Δx representing the infinitely small length of the particle

in any direction on the XY grid. Given that this balance of horizontal forces applies at any location and time, we can create a constant for this horizontal force from (5) as:

$$H = cos(\theta(x))T(x). \tag{6}$$

This constant applies to any position on the XY mesh grid. Our vertical forces have a similar representation with respect to the angle θ to represent the vertical forces, taking use of (6) to define the relation with the vertical forces:

$$T(x)\sin(\theta(x)) = H\tan(\theta(x)). \tag{7}$$

With this, we can define the infitely small displacement of our particle as a central difference in horizontal forces between the two sides to represent a relationship with our vertical forces. Thus we can represent:

$$tan(\theta(x)) = \frac{\partial u}{\partial x}(x,t). \tag{8}$$

With this relation we can provide a substitution into (4) after performing another central-difference approximation with the relation to horizontal forces, we can represent the sum of all forces as Δx infinitely approaches 0:

$$\frac{\partial^2 u}{\partial t^2}(x,t) = \alpha\frac{\partial^2 u}{\partial x^2}(x,t). \tag{9}$$

With α representing the constant ratio between horizontal forces and density of the wave as described in (2). Now that we have two-second order relationships, we can use a central difference approximation for multiple time intervals after defining two initial conditions and boundary conditions for the perimeter of the unit circle. For this model we will analyze the displacement of the unit circle over an interval of 5 s.

3.2 Boundary Conditions

3.2.1 The Heat Equation

The heat equation model allows for two types of boundary condition known as Dirichlet and Neumann boundary conditions. The easier of the two is the Dirichlet condition which states that

$$u(x,y,t) = g(x,y,t), \text{ if } (x,y) \text{ is on the boundary.} \tag{10}$$

For this to be valid, it is also assumed that $g(x, y, t)$ is known. For this model, it was set to a constant.

The Nuemann boundary condition then relates to the heat flux at the boundary of the shape. It states that a known amount of flux is leaving or entering the shape at specified points (x, y) on the boundary. To accomplish this using the discretized time stepping approach, an artificial temperature is set just outside the boundary so that the temperature difference ensures the proper amount of heat flux flows. These artificial temperatures can be computed using the following equations based on where the outside of the shape is located. N represents the intended heat flux in centimeter-gram-second (CGS), and the variables x_0 and y_0 represent the center location of the circular arc the boundary condition is located at. Through this boundary condition, it is also possible to specify that an edge is perfectly insulated by setting the flux to 0.

$$u_{(x+1,y,t)} = u_{(x-1,y,t)} - \frac{2N(x - x_0)\Delta x}{\sqrt{(x - x_0)^2 + (y - y_0)^2}}, \tag{11}$$

$$u_{(x,y+1,t)} = u_{(x,y-1,t)} - \frac{2N(y - y_0)\Delta x}{\sqrt{(x - x_0)^2 + (y - y_0)^2}}, \tag{12}$$

$$u_{(x-1,y,t)} = u_{(x+1,y,t)} + \frac{2N(x - x_0)\Delta x}{\sqrt{(x - x_0)^2 + (y - y_0)^2}}, \tag{13}$$

$$u_{(x,y-1,t)} = u_{(x,y+1,t)} + \frac{2N(y - y_0)\Delta x}{\sqrt{(x - x_0)^2 + (y - y_0)^2}}. \tag{14}$$

3.2.2 The Wave Equation

With a second-order partial differential equation, we need to define the initial conditions for both the base and first-order models. For our experiments, we will follow two apparatus, one with and one without a p-norm calculation that affects the boundary of the unit circle.

For a model of the standard unit disc (Model 1), our initial conditions shall be as follows:

$$u(x, y, 0) = \frac{x^2 + y^2}{10}, \frac{\partial u}{\partial t}(x, y, 0) = 0 \tag{15}$$

with the boundary condition that $u(x, y, t) = 0.1$ if $x^2 + y^2 = 1$.

For a model of the standard unit disc at multiple defined p norm values (Model 2), our initial conditions shall be as follows:

$$u(x, y, 0) = \frac{||(x, y)||_p}{10}, \frac{\partial u}{\partial t}(x, y, 0) = 0 \tag{16}$$

with the boundary condition that $u(x, y, t) = 0.1$ if $|(x, y)||_p = 1$

For both models, since the points on the mesh are defined in a rectangular matrix, and the boundary conditions are in terms of the boundary of a circle, we will use an approximation to define the points that are in and outside of the considered area. Using a precision of 0.05 m, the distance formula is calculated at every point to determine how close each point is to the unit circle boundary of $r = 1$. If the point on the XY grid is determined to be outside the border, it will permanently have the displacement of $z = 0.1m$ for all time.

With initial conditions and boundary conditions defined, we set up an iterative central-difference approximation calculation method. To simplify display, we define two constants used in our method:

$$R_x = \frac{\alpha \Delta t^2}{\Delta x^2}, \tag{17}$$

$$R_y = \frac{\alpha \Delta t^2}{\Delta y^2}. \tag{18}$$

We substitute (17) and (18) into our finalized central difference method computed from (9):

$$\frac{\partial^2 u}{\partial x^2} = \frac{u(x, t + \Delta t) - 2u(x, t) + u(x, t - \Delta t)}{\Delta t^2}. \tag{19}$$

Given that we know the first two time increments of our central difference method since, for both of our models, our initial first-order derivative is 0, we can create an iteration for the next time increment in terms of both the current and previous instance of the displacement at any x_i and y_j:

$$\begin{aligned} u_{x,y,t+1} =& 2u_{x,y,t} - u_{x,y,t-1} \\ &+ R_x(u_{x+1,y,t} - 2u_{x,y,t} + u_{x-1,y,t}) \\ &+ R_y(u_{x,y+1,t} - 2u_{x,y,t} + u_{x,y-1,t}). \end{aligned} \tag{20}$$

3.3 Stability and Constraints

3.3.1 The Heat Equation

Due to the nature of the time stepping solution method, it is critical to have a small enough discretization to ensure both the validity and stability of the results. A Δt value that is too large will result in divergent instability while too large of Δx and Δy values will give results that do not match the intended geometry. For the thermal diffusion model, a maximum Δt value can be computed using the following equation.

$$\Delta t < \frac{c\rho\Delta x^2\Delta y^2}{2k(\Delta x^2 + \Delta y^2)}. \tag{21}$$

To ensure convergence, the Δt value was set to 0.9 times the maximum for the model.

The heat equation also requires that all edges of the shape are given a boundary condition and that the initial condition was set before running the model.

3.3.2 The Wave Equation

In order to ensure stability in the time-stepping, the time increments of the model at which each frame will be computed at shall follow the rule:

$$\Delta t \leq \Delta x\sqrt{\alpha}. \tag{22}$$

4 Model Validation and Results

4.1 Validation Process

4.1.1 The Heat Equation

The thermal diffusion model was hard coded to calculate the heat transfer for the shape shown in exercise 1 in chapter 11.4 of Holder and Eichholz (2019). This shape is also described as a pill shape. It consists of two semicircles facing each other with diameters of 10 cm that are 10 cm apart. The shape initially starts at a uniform 60 °C. The two straight line edges are kept at 60 °C through the Dirichlet boundary condition while the curved edges have

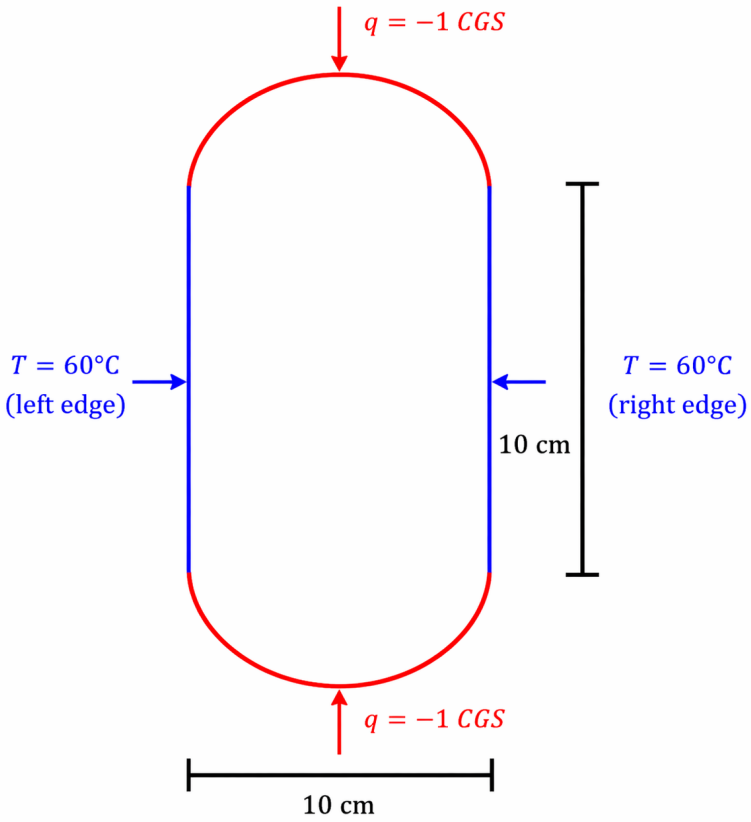


Figure 1: Geometry of shape used in heat transfer model.

a specified -1 CGS heat flux. This can be seen in figure 1. Each simulation will be run for a total of 2 seconds.

The results of the thermal diffusion model were not tested against real world data or known temperature states for the given geometry, but rather, evidence was found for the model based off how common materials are known to spread heat according to their physical properties. For example, copper is known for being a good conductor of heat while not retaining a lot of that heat in the process. The physical properties used for copper were $k = 0.94$, $c = 0.092$, and $\rho = 8.96$. We would expect the model to have a semi-steep temperature gradient between areas with heat coming into the shape and areas with heat leaving. The results for the model being run with copper

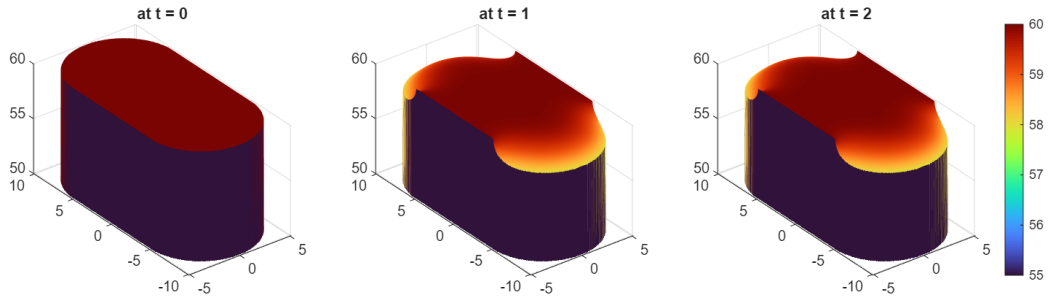


Figure 2: Heat transfer of copper.

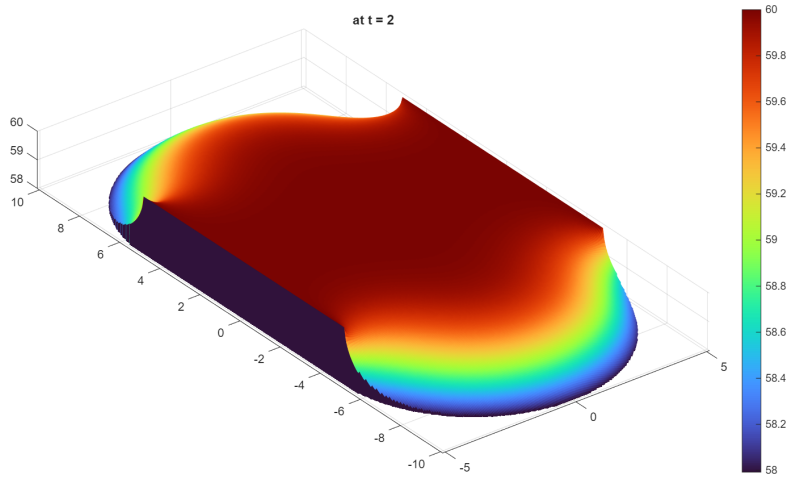


Figure 3: Detailed view for copper.

parameters can be seen in figure 2. The following figure, figure 3, shows the final state in greater detail. From this, the model appears to be working correctly with regard to how heat is spreading through the material.

Note that for all thermal diffusion figures, the z-axis only shows temperatures from 50 to 60 °C and the color bar only ranges from 55 to 60 °C, or an even more limited range for the detailed follow-up plots.

To ensure the model accurately captures the intended geometry, a study was done to see the effect of Δx and Δy on temperature in a region. To do this, the minimum temperature of the shape was plotted vs Δx and Δy, which are equal to each other. These trials were also run with copper as the material. The results of this can be seen in figure 4. Through applying a best fit line through the data, we can approximate what the results would be

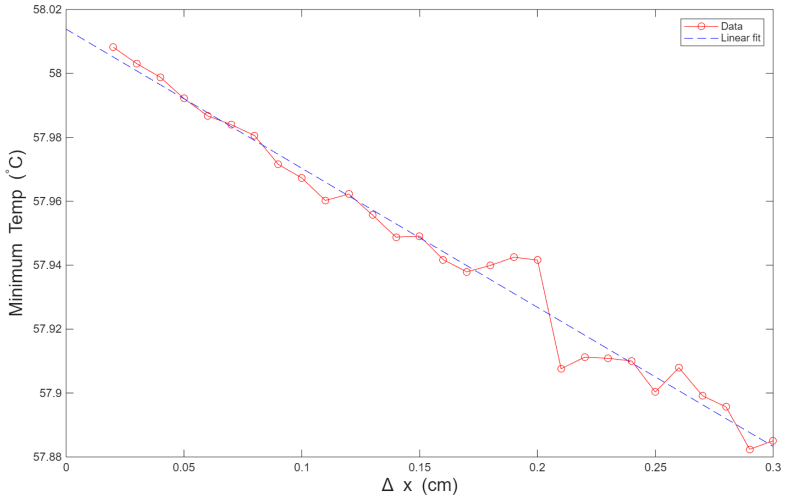


Figure 4: Minimum temperature vs Δx for copper.

if $\Delta x = 0$. This would be representative of a perfect model. To expand this "perfect model" to other materials, the intercept of this line must be made a function of α. A best fit line was then applied to data consisting of the intercept vs α for various materials, shown in figure 5. This fit equation is then used to determine the "perfect model" minimum temperature for any material. That equation is shown below:

$$u_0 = -1.43\alpha + 59.73. \tag{23}$$

Using this to calculate values for minimum temperature, we can compare against the model's output to give an estimation of the error for different values of Δx and Δy. A table containing this data can be found below. For every case calculated here, the error was within 0.2 °C when Δx was set to 0.05 cm. Note that this error is specific to only the specified geometry, initial state, and boundary conditions.

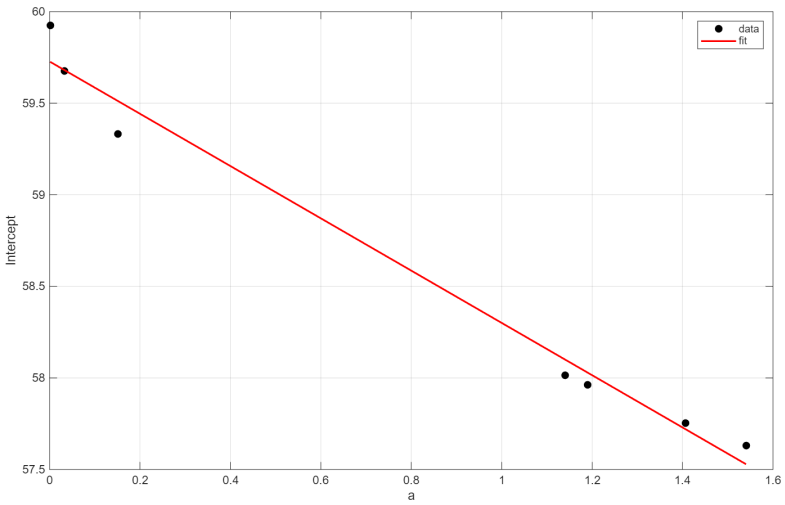


Figure 5: Intercept vs α for various materials.

Material	$\Delta x \approx 0$	$\Delta x = 0.05$ cm	Error
Steel	59.51	59.33	0.18
Copper	58.10	57.99	0.11
Water	59.73	59.93	0.20
Hydrogen	57.72	57.73	0.01
Helium	57.53	57.61	0.08
Diamond	58.03	57.96	0.07
Aerogel	59.68	59.68	0.00

Table 1: Comparison of minimum temperatures.

4.1.2 The Wave Equation

The validation process of the wave equation model was to plot the second-order difference method over multiple increments of time to check for standard, visual wave response.

Figures 6 and 7 depicts how the unit circle will be rendered as a 3-dimensional graphic for various conditions and times.

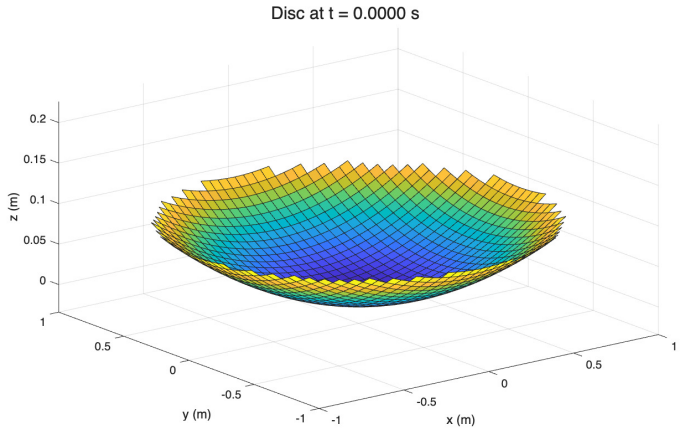


Figure 6: The initial position of the unit disc plot.

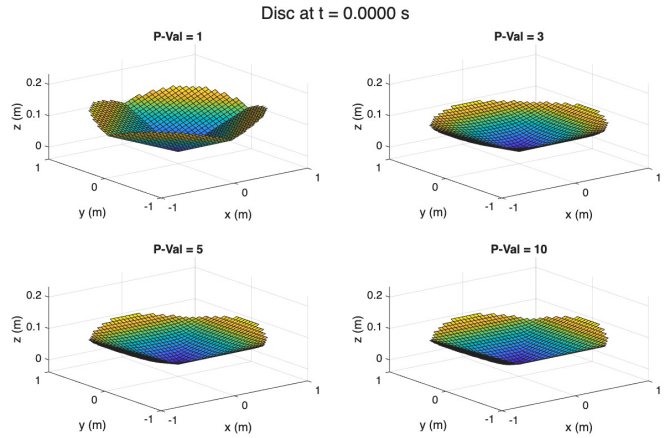


Figure 7: The initial position of the unit disc plot at different p-norm values as indicated above each subplot.

For both experiments, the following parameters were used in the computation and plotting. The constant $\alpha = 2.0$ was used through and the displacement of the increments on the mesh grid $\Delta x, \Delta y$ were set to 0.05 m. A total modeling time of 5 s was used, and the increments in time were determined by the constraint described in (22).

4.2 Experiments

4.2.1 The Heat Equation

The experiment done for heat transfer of the pill shape involves determining the material for the most uniform heat profile after 2 seconds. Several materials were chosen for testing due to certain physical characteristics such as extreme values for k, c, and ρ. The following table will include these materials and their properties. Their calculated α values can also be seen here. Note that these values are not exact, in reality they change with temperature but here they are modeled as constants. These constants are approximated from a variety of online tables and sources.

Material	k (cal/(cm·s·°C))	c (cal/(g·°C))	ρ (g/cm^3)	α (cm^2/s)
Steel	0.13	0.11	7.8	0.1515
Copper	0.94	0.092	8.96	1.1403
Water	0.0014	1	1	0.0014
Hydrogen	0.00043	3.4	0.0000899	1.4068
Helium	0.00034	1.24	0.000178	1.5404
Diamond	5.0	1.2	3.5	1.1905
Aerogel	0.00002	0.2	0.003	0.0333

Table 2: Thermal properties of materials.

The optimal material for most uniform heat profile depends only on the α value due to how the heat transfer in modeled based on equation 1. It can then be determined that water will have the most consistent heat profile due to a combination of its high heat capacity, low thermal conductivity, and average density. The results of water subject to the same conditions as before can be seen in figures 8 and 9.

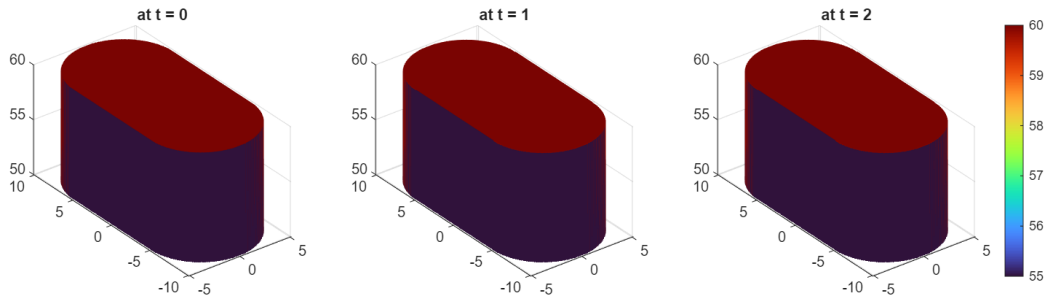


Figure 8: Heat transfer of water.

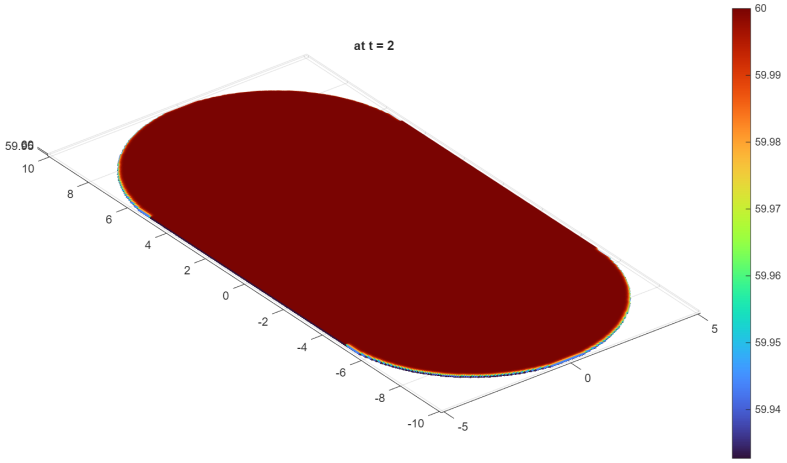


Figure 9: Detailed view for water

The temperature of water after 2 seconds of runtime is so uniform that it is hard to see any difference in the above figures. The minimum temperature found on the shape was only 59.93 °C, making for a maximum temperature difference of 0.07 °C. A table detailing the average, minimum, and difference between the maximum and minimum temperatures for all the materials can be found below. For a more uniform temperature, a small difference between the minimum and maximum is desired.

Material	Min (°C)	Avg (°C)	Diff (°C)
Steel	59.33	59.95	0.67
Copper	57.992	59.62	2.008
Water	59.93	59.9992	0.07
Hydrogen	57.73	59.54	2.27
Helium	57.61	59.50	2.39
Diamond	57.96	59.61	2.04
Aerogel	59.68	59.98	0.32

Table 3: Temperature statistics for different materials.

4.2.2 The Wave Equation

With the standard unit-disc, the model showed a flush, uniform change in position over the 5-second interval as depicted in the following Figures.

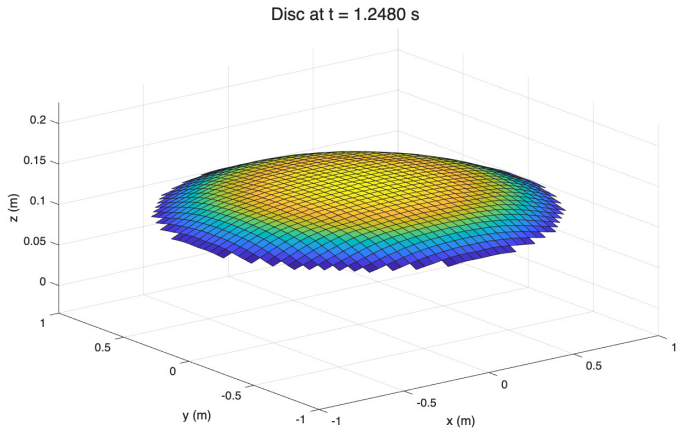


Figure 10: The the unit disc plot at t = 1.2480 s. A mound begins to grow in the center of the disc.

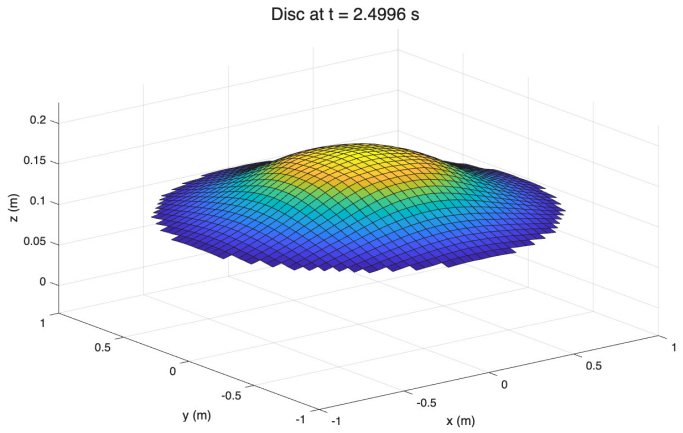


Figure 11: The the unit disc plot at t = 2.4996 s. The mound now appears to be approaching its peak, with the surrounding area concaving downward.

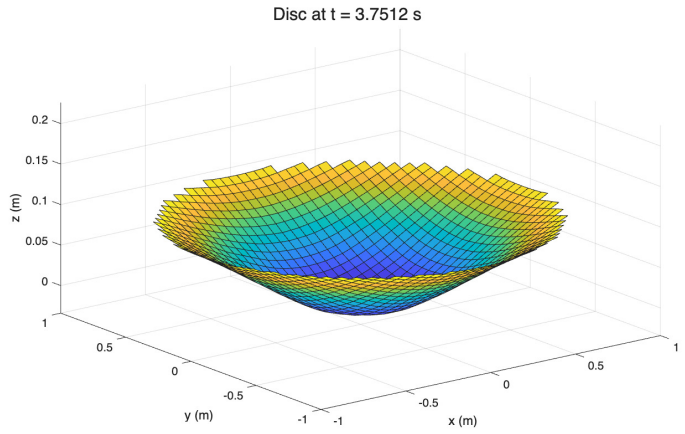


Figure 12: The the unit disc plot at t = 3.7512 s. The mound has now inversed.

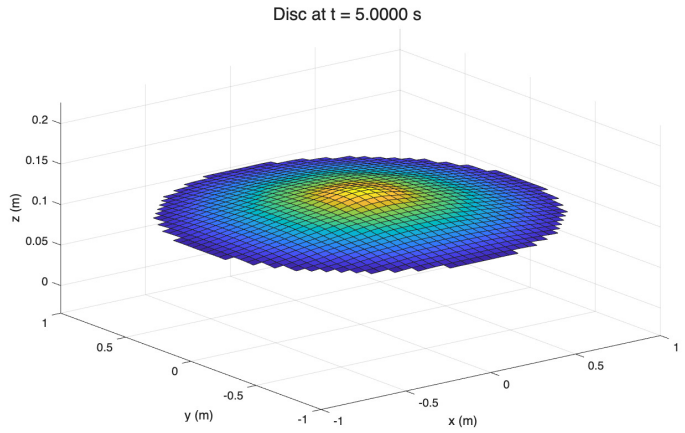


Figure 13: The the unit disc plot at t = 5.0000 s. The mound has re-approached the height of the outer boundary condition.

This wave equation reflects that of an object being dropped into a pool of water, as the a flexible fluid can have such properties and react in an oscillatory, uniform manner.

The p-norm varient of the unit disc focused on four values of the p-norm; $p = 1, 3, 5$, and 10. Altering this p-norm changes the outside shape of the boundary condition, thus affecting how the initial conditions play a role on the subsequent and future iterations of time on the model. In this experiment, we wanted to approximate the shape of the unit disc as $p \to \infty$. In an ideal model, the boundary condition of the unit disc would appear as a square. To replicate a real-world version, we used $p = 10$, as p-vals beyond this point have marginal change to the outer boundary layer. P-norm value $p = 1$ is known to have a unique affect on the plot compared to any other value, so it will be used simply as a means of comparison with the other plots.

Like the standard unit circle disc, the p-norm values were plotted in a uniform change over the 5 second interval, as depicted in the subsequent figures.

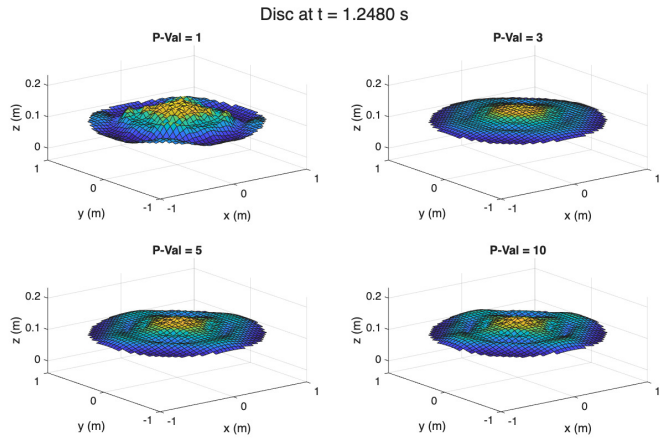


Figure 14: The the unit disc plot with multiple p-values at t = 1.2480 s. Multiple rifts appear on our "standard" p-values, while p-val = 1 has more jagged rifts.

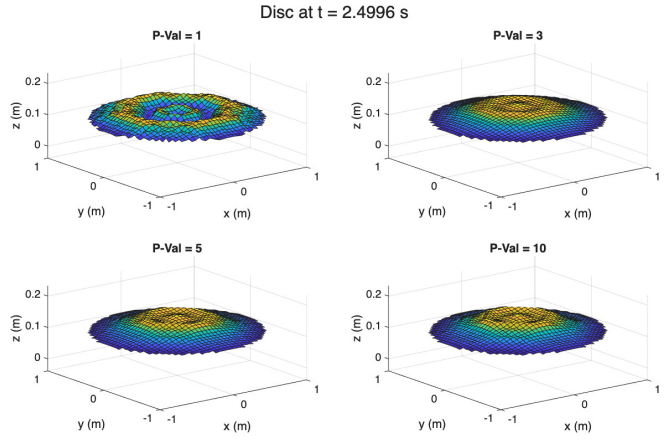


Figure 15: The the unit disc plot with multiple p-values at t = 2.4996 s. Higher mounds of the rifts have appeared, but our p-val 1 plot has the mounds spread further apart.

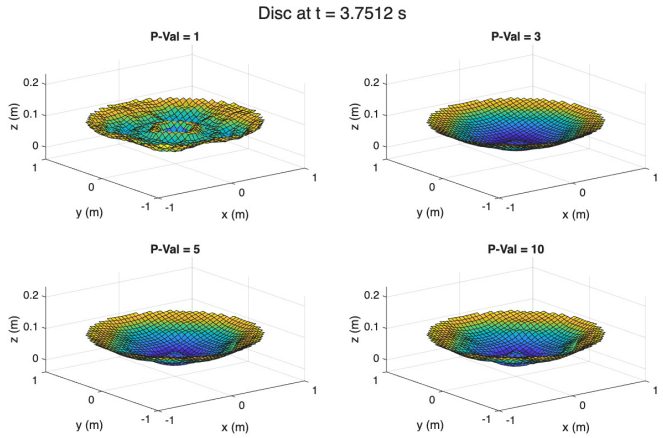


Figure 16: The the unit disc plot with multiple p-values at t = 3.7512 s. All of our plots have a "dipped" bowl shape, but our p-val 1 plot has multiple ripples through the bowl shape, as well as an upward-facing center.

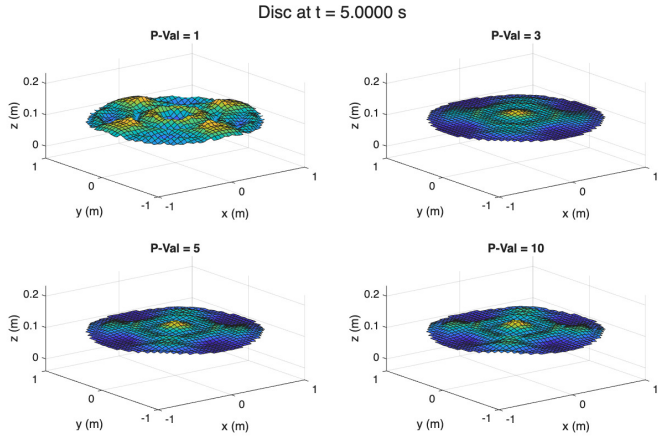


Figure 17: The the unit disc plot with multiple p-values at t = 5.0000 s. The plots have returned to the boundary layer's height, but have a small peak in the center that gradually extends upward as the p-value increases. For p-value 1, the affect appears to be the opposite, with a bowl-shape in the center.

As our p-values increase, the z-axis has more sensitive adjustments at various time increments, such as the more conditioned positions along the diagonals on the XY grid in Figure 17. The increased p-value for each computation creates more rigid affects along the outside edge, such as the fourth subplot in Figures 14 and 16. For our $p = 1$ variant, we have an extreme difference in anticipated response and is much more turbulent in behavior.

5 Conclusion

After experiments, both models showed reliability in consistent data and graphical modeling solutions. The thermal diffusion model was able to calculate heat transfer across a specified shape for various materials and the wave model was able to simulate how waves propagate across a unit disk at various p-norms. These models can be used for future custom applications in various applicable working industries.

References

[1] Allen Holder and Joseph Eichholz. *An Introduction to Computational Science*, volume 278. Springer Nature Switzerland AG, 2019.